Varun Babbar

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Summary

ML researcher (CS PhD @ Duke, 1st Class Cambridge undergrad) with 5+ years of experience across industry and academia. First-author papers at NeurIPS, ICML **(1 Oral, top 1%; 1 Spotlight, top 2%)**, IJCAI **(Oral)**, and JMLR. Specialized in interpretable machine learning on tabular data (decision trees, linear models, large-scale foundation models), with an adjacent record of success in applied AI in industry **(1 patent, 1 pending patent)**.

Education

Duke University — *Durham, NC, USA*
PhD Candidate, Computer Science — *August 2023 – Expected May 2027*
Advisor: *Cynthia Rudin*

University of Cambridge — *Cambridge, United Kingdom*
BA + MEng in Information and Computer Engineering — *2018-2022*
Advisors: *Adrian Weller (MBE), Umang Bhatt* — *1st Class (BA), Distinction (MEng) - **Top 5% of class***

Research Experience

Duke Interpretable Machine Learning Lab — Aug 2023 – Current
Advisor: Dr Cynthia Rudin

- Designed fast algorithms to find near-optimal decision trees and generalized additive models on noisy, high-dimensional data. 3 first-author **ICML papers** (1 **Spotlight**, 1 **Oral**) + 1 first-author **NeurIPS paper**.
- Built a toolbox for analyzing distribution shifts in tabular and vision data (**JMLR**, ∼ 17% **acceptance rate**).
- Currently working on interpretability of tabular foundation models and uncovering regimes where classical interpretable models perform better.

University of Cambridge: Department of Engineering — Sept 2021 – July 2022
Advisor: Dr Adrian Weller (MBE), Dr Umang Bhatt

- Developed human-aligned uncertainty quantification methods for deep learning models with applications to model oversight and intervention prioritization. **IJCAI Oral** (**top** 3%) + **NeurIPS** workshop papers.

University of Cambridge: Department of Computer Science — June 2020 – Sept 2020
Advisor: Dr Rafal Mantiuk.

- Developed state-of-the-art perceptual loss functions for **image restoration tasks** (super-resolution, deblurring, denoising).
- Co-author of paper accepted at IEEE Winter Conference on Applications of Computer Vision **(WACV '22)**.

Industry Experience

Research Scientist Intern at Capital One — *New York, United States*
June 2026 - August 2026

- Working on pre-training of foundation models for prediction tasks on tabular data. *Paper submitted to NeurIPS 2026 workshops.*

Applied Scientist Intern at Zillow — *New York, United States*
May 2025 – Aug 2025

- Developed a multimodal representation learning algorithm for generating home embeddings that integrates visual, tabular, and graph features.

- Trained on over 10 million home listings across the US, with evaluations showing state-of-the-art performance on internal recommendation benchmarks.

AI Scientist at JP Morgan Chase and Co *London, United Kingdom*
Sept 2022 - July 2023

- Worked on evaluation of large language models for source code understanding.
- Won a firmwide generative AI hackathon, published a paper at a SWE conference, developed a unit test quality assessment tool that saved the firm ~ $2 million, contributed to open source, and a pending patent.

AI Scientist Intern at JP Morgan Chase and Co *London, United Kingdom*
June 2021 - Sept 2021

- Developed a federated deep learning algorithm robust to distribution shifts in client datasets. Outcome: a patent and a paper accepted at SPIE Medical Imaging 2022.

Academic Awards and Honours

Graduate School Fellowship @ Duke University *Aug 2023*
JP Morgan Innovation Week AI Hackathon *(Runner Up)* *June 2023*
MEng Project Prize *(For a top ranked MEng Project @ Cambridge)* *July 2022*
Ruth Hendry Year Prize *(For exemplary academic performance)* *June 2022*
Foundation Scholar (2x) *(For 1st class performance in exams)* *Oct 2021, Oct 2022*
The James and Jean Bennet Prize *(For exemplary academic performance)* *June 2022*
Hawkes' Trust Award *(For exemplary academic performance)* *March 2021*
HackCambridge, Oxford Hackathon, HackBrunel *(Winner)* *Jan 2022, Nov 2018, Oct 2019*
World Topper in IB School Exams *(100th percentile out of ~170000 students)* *May 2018*

Conference, Industry, and Academic Presentations

ICML 2026, Seoul, South Korea July 2026
NeurIPS 2025, San Diego, USA December 2025
ICML 2025, Vancouver, Canada July 2025
New York University, Responsible Data Science Course April 2025
Responsible AI Symposium, Duke University Feb 2025
Duke University, Guest Lecture, MEng in AI Program Sept 2024, Feb 2025
JP Morgan, Innovation Week Speaker Series, London, UK June 2023
NeurIPS 2022 Workshop on Human-in-the-Loop Learning, Virtual Dec 2022
JP Morgan, Applied Innovation in AI Seminar, London, UK Nov 2022
International Joint Conference on AI (IJCAI 2022), Vienna, Austria July 2022
ICML Workshop on Distribution Free Uncertainty Quantification, Virtual July 2022
University of Cambridge, Department of Engineering July 2022

Service

Research Mentorship: Michael Thomas, Allen Yao, Chris Li - CS+Math Undergrads @ Duke Jan 2025-Current
Yixiao Wang - MS in Statistics @ Duke June 2025-Current
Program Committee: Interpretable AI Workshop @ NeurIPS Dec 2024
Reviewer: ECAI 2023, WANT@ICML 2024, TheWebConf2023, Journal of the ACM, AISTATS 2025, ICML 2026, NeurIPS 2026

Skills

ML/Research: PyTorch, Transformers, optimization (discrete, stochastic, constrained), statistics (Bayesian, uncertainty quantification)
Engineering: Python (NumPy, Pandas, Scikit-Learn, HuggingFace, W&B), Git, Bash, LaTeX